

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	User Input Module	Accepts soil and environmental parameters (N, P, K, Temperature, Humidity, pH, Rainfall).	User Interface	Done	High
2	Data Preprocessing	Cleans and prepares the dataset for machine learning.	Data Processing	Done	High
3	K-Means Clustering	Groups similar agricultural conditions for data analysis.	Machine Learning	Done	Medium
4	Logistic Regression Model	Predicts the most suitable crop using input parameters.	Machine Learning	Done	High
5	Crop Recommendation	Generates the recommended crop based on model prediction.	Prediction Module	Done	High
6	Model Saving	Saves the trained model using Pickle for future use.	Model Management	Done	Medium
7	Flask Web Application	Integrates the trained model with the web interface.	Backend	Done	High
8	Result Display	Displays the predicted crop recommendation to the user.	User Interface	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Input Form, Result Display
2	Backend / Logic	2	Flask Backend, Crop Recommendation Logic
3	Database / Storage	1	Pickle Model, csv dataset
4	API / Integration	1	Flask Model Integration
5	Security / Authentication	0	Not Implemented